

Function 2: CER Accounting

CER calculation, unit-level query, invalid energy-type rejection, and block-level confirmation.

CER Accounting Block Summary

Block-level evidence decoded from Fabric ledger

HEIGHT CHANGE 4 -> 5	BLOCK NUMBER 4	CHANNEL mychannel	TIMESTAMP 2026-05-27T07:09:12.071Z
CURRENT BLOCK HASH 0j3YU7AACX2733+4nS4kZf9mxdC44w2mk9YybJBHT0M=		PREVIOUS BLOCK HASH 7VcNQ8imvHXBPC3Hcea19kg3cCQVvUQFv0t7UaYjVKo=	
TRANSACTION ID 0cbc91e5ccd6babfe3ca4ba43f574d1003c0c7725020be8b84cd35a79a53617f		DATA HASH imqpI/KrFDchg0CQRI/qK9FavoRPgwF1zjl34Dm9C9Y=	

CER Input Evidence

input provenance

INPUT FILE energy_valid.json	RECORDS PARSED 2,000 units	CHAINCODE FUNCTION EmissionReduction	VALIDATION INPUT energy_invalid_type.json
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CER SDK Output Payload

emission_reduction_valid.log

PARSED RECORDS 2,000	TOTAL CER 48.56 tCO2e	GAS / ELECTRICITY RECORDS 1,000 / 1,000	FIGURE OUTPUT first 6 JSON records
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payload from emission_reduction_valid.log

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{"unit_id":1,"accounting_time":"2025-11-15","energy_type":"gas","emission_reduction":33600}
{"unit_id":2,"accounting_time":"2025-11-15","energy_type":"gas","emission_reduction":29400}
{"unit_id":3,"accounting_time":"2025-11-15","energy_type":"gas","emission_reduction":25200}
{"unit_id":4,"accounting_time":"2025-11-15","energy_type":"gas","emission_reduction":21000}
{"unit_id":5,"accounting_time":"2025-11-15","energy_type":"gas","emission_reduction":16800}
{"unit_id":6,"accounting_time":"2025-11-15","energy_type":"electricity","emission_reduction":35040}
... 1994 additional records omitted from figure view
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CER Query and Validation Results

validation checks

UNIT-LEVEL QUERY Unit 1 = 0.0336 tCO2e; QueryEmissionResult confirmed	INVALID INPUT REJECTION Rejected invalid energy_type = steam; expected failure captured
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